Project Documentation Report

Streamlining Ticket Assignment for Efficient Support Operations

Date	5 April 2026
Team ID	SWTID-2026-8642
Project Name	Streamlining Ticket Assignment for Efficient
Maximum Marks	2 Marks

Team Members

S.No	Name	RegNo
1	Lakshmanan S	812823205041
2	Gokulakannan S	812823205015
3	Jegan R	812823205028
4	Aravind A	812823205004
5	Bavithran M	812823205008

1. INTRODUCTION

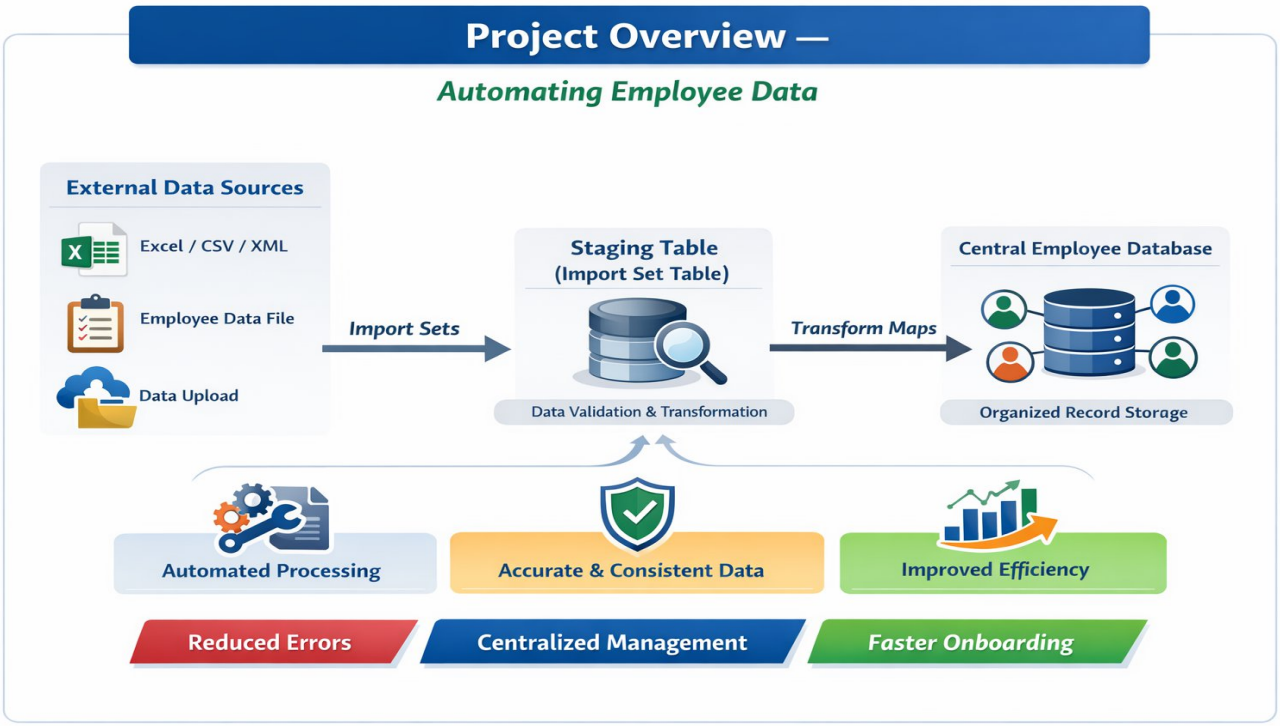
Project Purpose:

In today's fast-paced digital environment, customer support teams handle a large number of tickets every day. Efficiently assigning these tickets to the right support agents is crucial for quick resolution and customer satisfaction. Manual ticket assignment can be time-consuming, error-prone, and inefficient, especially as the volume of requests grows. Therefore, there is a need for a streamlined and automated approach to manage ticket distribution effectively.

The purpose of this project is to improve the efficiency of support operations by automating and optimizing the ticket assignment process. By ensuring that tickets are directed to the most suitable agents based on factors such as expertise, workload, and priority, the system aims to reduce response time, enhance productivity, and improve overall customer experience.

Project Overview:

This project focuses on developing a system that automatically assigns incoming support tickets to appropriate team members. The system analyzes ticket details and matches them with agent skills and availability. It helps balance workloads, minimize delays, and ensure that issues are resolved quickly and accurately. Ultimately, the solution makes support operations more organized, efficient, and scalable.



Key Objectives & Expected Outcomes:

Key Objectives:

The main objective of the project *"Streamlining Ticket Assignment for Efficient Support Operations"* is to develop an effective and automated system for managing and assigning support tickets. The system aims to reduce manual effort by automatically categorizing and assigning tickets to the appropriate agents based on factors such as issue type, priority, and agent availability.

Expected Outcomes:

It is expected that the implementation of this system will significantly reduce manual effort and minimize errors in ticket assignment. The response time for handling customer issues will be improved, leading to faster resolution and enhanced operational efficiency. The system will also enable balanced workload distribution among support agents, preventing overloading and improving productivity.

2. IDEATION PHASE

Problem Statement Definition

Our Problem Statement:

In many organizations, employee data is managed manually using spreadsheets or multiple disconnected systems, leading to issues such as data duplication, errors, inconsistency, and delays in updating records. This makes it difficult to maintain accurate employee information and slows down important processes like onboarding and data management. Therefore, there is a need to develop an automated system using ServiceNow that can efficiently import, validate, and manage employee data in a centralized and structured manner, ensuring accuracy, consistency, and improved operational efficiency.

Problem Statements Identified

PS-1

Problem Statement (PS): PS-1
I am (Customer): A Customer Support Agent
I'm trying to: assign incoming support tickets to the appropriate team or agent efficiently
But: the current process is manual, time-consuming, and sometimes results in incorrect assignment of tickets
Because: there is no automated workflow to categorize and route tickets based on priority, issue type, and agent availability
Which makes me feel: overwhelmed and inefficient due to handling multiple tickets manually

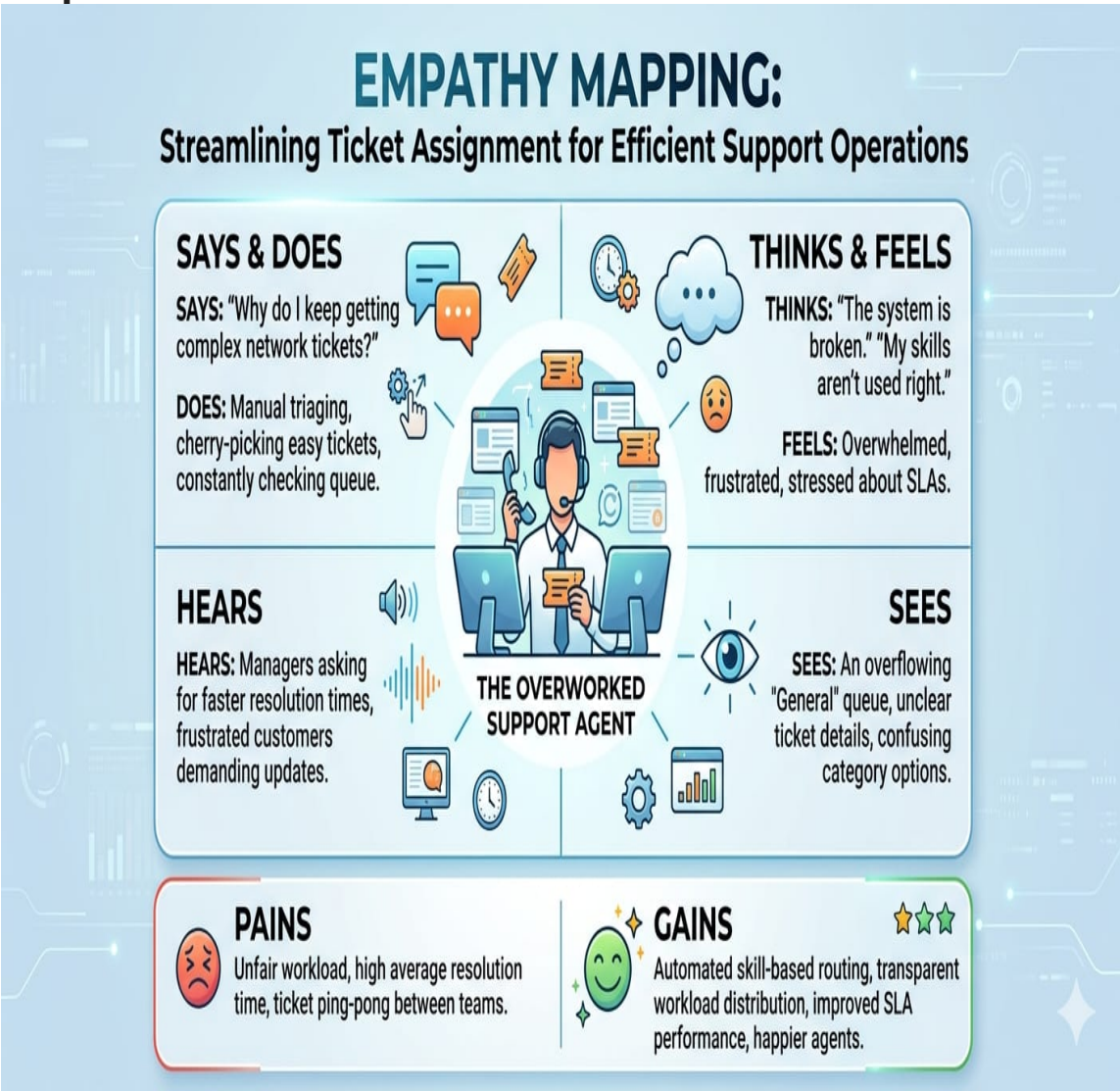
PS-2

Problem Statement (PS): PS-2
I am (Customer): A Customer / End User
I'm trying to: get my support issues resolved quickly and accurately
But: my ticket is often delayed or assigned to the wrong department
Because: the system does not have proper ticket classification and intelligent assignment
Which makes me feel: frustrated and dissatisfied with the service

PS-3

Problem Statement (PS): PS-3
I am (Customer): A Support Manager / Organization
I'm trying to: ensure efficient handling and tracking of all support tickets
But: the system has delays, misrouted tickets, and uneven workload distribution among agents
Because: there is no proper automation and validation in the ticket assignment process
Which makes me feel: concerned about operational efficiency and customer satisfaction

Empathy Map:


EMPATHY MAPPING:
Streamlining Ticket Assignment for Efficient Support Operations
SAYS & DOES
SAYS: "Why do I keep getting complex network tickets?"
DOES: Manual triaging, cherry-picking easy tickets, constantly checking queue.
THINKS & FEELS
THINKS: "The system is broken." "My skills aren't used right."
FEELS: Overwhelmed, frustrated, stressed about SLAs.
HEARS
HEARS: Managers asking for faster resolution times, frustrated customers demanding updates.
THE OVERWORKED SUPPORT AGENT
SEES
SEES: An overflowing "General" queue, unclear ticket details, confusing category options.
PAINS
Unfair workload, high average resolution time, ticket ping-pong between teams.
GAINS
Automated skill-based routing, transparent workload distribution, improved SLA performance, happier agents.

Brainstorming and Idea Prioritization:

1. **Problem Statement**

   Automate employee data import into ServiceNow and maintain accurate records without manual entry.

2. Brainstorming

- Import: Scheduled Import Sets vs. REST API vs. Manual Upload
- Processing: Transform Maps vs. Business Rules
- Storage: sys_user Table vs. Custom Table

3. Prioritization

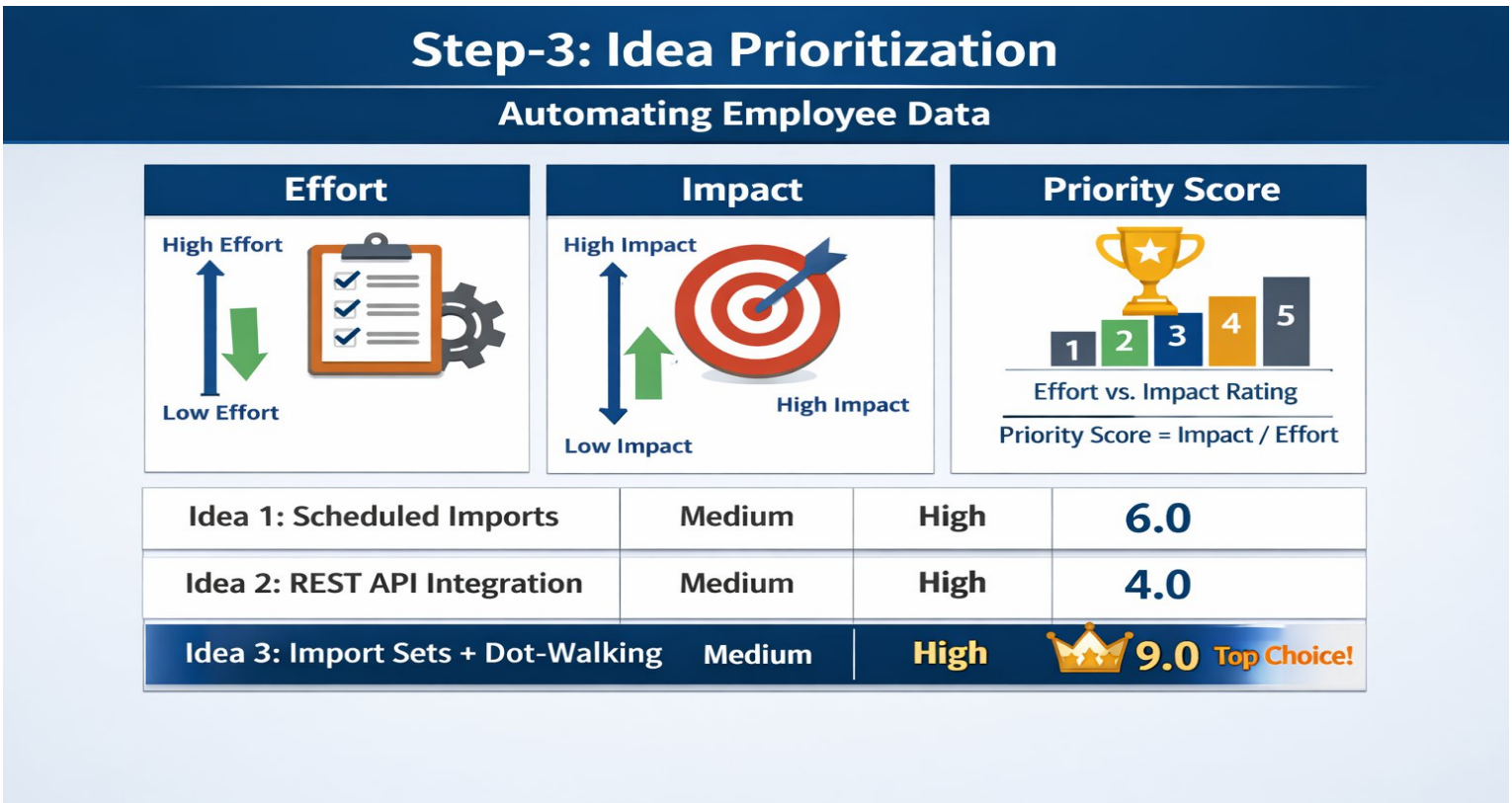
Selected Approach: Import Sets + Transform Maps + sys_user Table.

Chosen for better performance, accuracy, and ease of implementation.

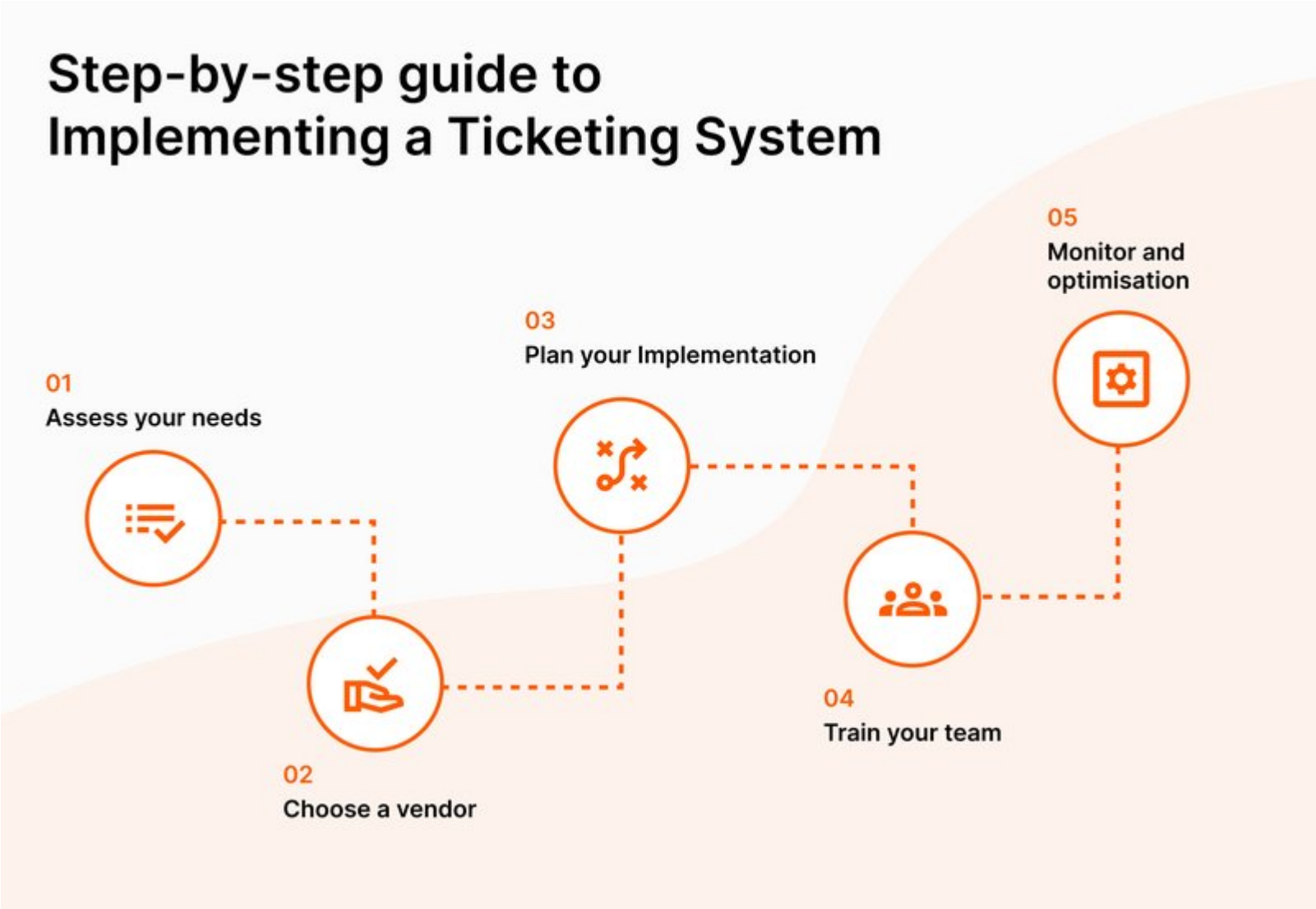
Step-1: Team Gathering, Collaboration and Select the Problem statement.

Step-2: Brainstorm, Idea Listing and Grouping

Step-3: Idea Prioritization



3. REQUIREMENT ANALYSIS



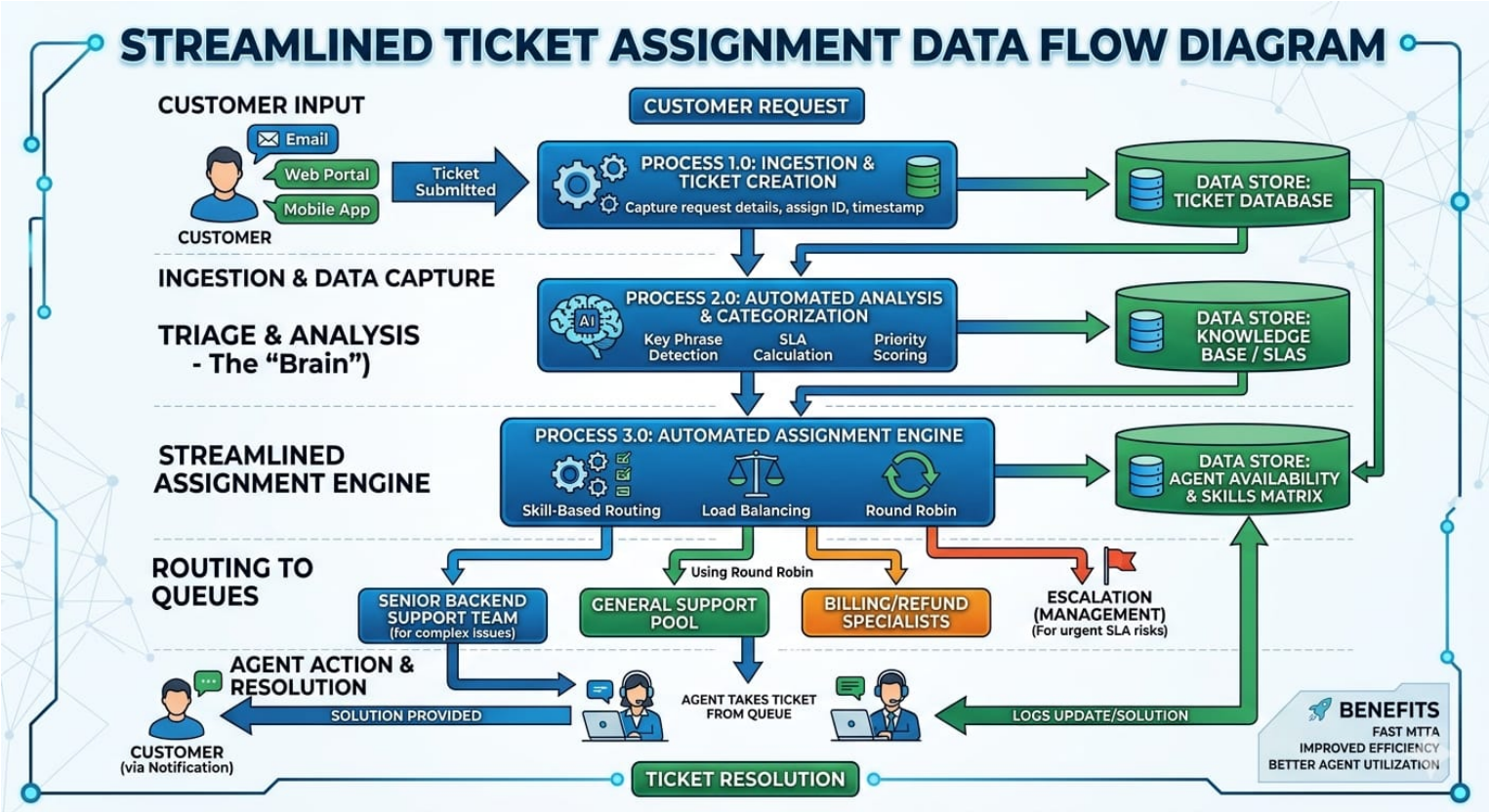
Data Flow Diagrams:

Data Flow Diagrams (DFD) for *Streamlining Ticket Assignment for Efficient Support Operations* represent how data moves through the support system in an organized way. When a user submits a ticket, the information flows into the system where it is validated and stored in a ticket database.

The system then analyzes the ticket details and automatically assigns it to the most appropriate support agent based on factors such as skill set, availability, and workload.

Once assigned, a notification is sent to the agent, and the ticket status is updated continuously as the issue is being resolved.

After completion, the final response is communicated back to the user. This flow of data helps reduce manual intervention, improves response time, ensures proper resource utilization, and enhances overall efficiency in support operations.



Technical Architecture:

The technical architecture of the ticket assignment system is designed to ensure efficient handling, processing, and routing of support tickets. The system follows a structured flow that begins with the user interface and ends with reporting and monitoring.

At the initial stage, the **Customer or End User** interacts with the system through a **Support Interface**, where they submit support tickets by providing details such as issue type, description, and priority. This interface acts as the entry point of the system.

Once the ticket is submitted, it moves to the **Data Processing and Categorization module**, where the system analyzes the ticket details. In this stage, the ticket is classified based on predefined categories such as technical issue, network problem, or software request. Priority levels (High, Medium, Low) are also assigned.

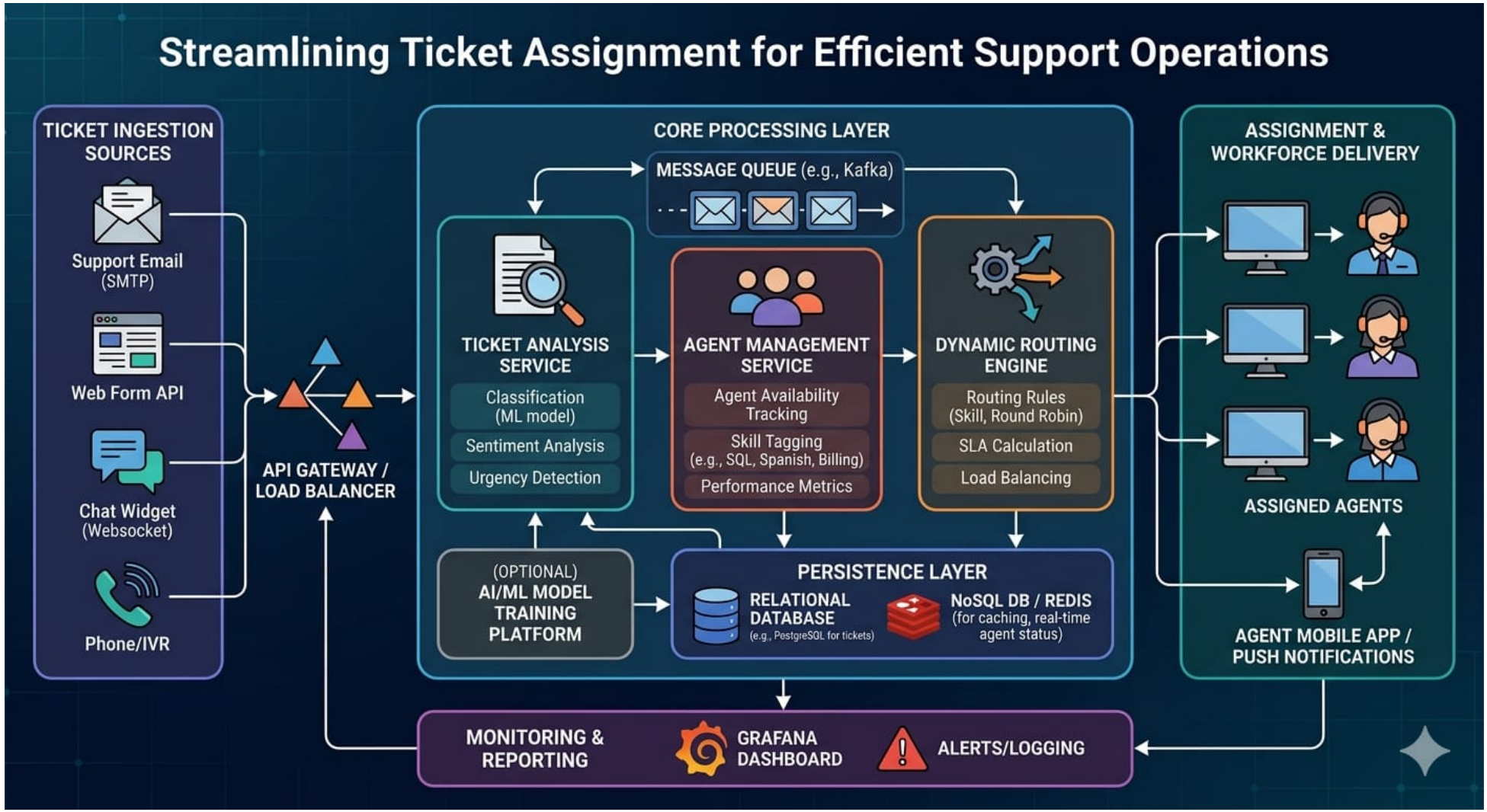
Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interface for users (customers, agents, admin) to interact with the system (Web App / Dashboard)	HTML, CSS, JavaScript / Angular Js / React Js etc.
2.	Application Logic-1	Handles user registration, login, and authentication processes	Java / Python
3.	Application Logic-2	Ticket creation, categorization, and priority assignment logic	Python / Node.js
4.	Application Logic-3	Automated ticket assignment using rules or AI-based decision making	Machine Learning / Rule-based Engine
5.	Database	Stores user data, ticket details, status, and logs	MySQL, NoSQL, etc.
6.	Cloud Database	Cloud-based storage for scalability and remote	Firebase / AWS RDS / MongoDB Atlas

		access	
7.	File Storage	Stores attachments such as screenshots, documents in tickets	AWS S3 / Local Storage
8.	External API-1	Email and notification service for alerts and updates	SMTP / SendGrid API.
9.	External API-2	Integration for authentication or third-party login (Gmail, LinkedIn)	Google API / LinkedIn API
10.	Machine Learning Model	Predicts ticket priority and assigns to appropriate support agents	Classification Model (AI/ML)
11.	Infrastructure (Server / Cloud)	Deployment and hosting of application	AWS / Azure / Google Cloud / Local Server

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used for building the web-based ticket management system	React JS (Frontend), Node.js / Django (Backend)
2.	Security Implementations	Secure authentication, data protection, and access control for users and admin	JWT Authentication, SHA-256 Encryption, HTTPS, Role-Based Access Control (RBAC)
3.	Scalable Architecture	System designed to handle increasing users and tickets efficiently using modular architecture	3-Tier Architecture (UI, Logic, Database), Microservices (optional), Docker
4.	Availability	Ensures system is always accessible with minimal downtime	Cloud Hosting (AWS/Azure), Load Balancer, Distributed Servers
5.	Performance	Optimized for fast response and handling multiple requests simultaneously	Caching (Redis), CDN, Optimized Database Queries


Streamlining Ticket Assignment for Efficient Support Operations
TICKET INGESTION SOURCES
Support Email (SMTP)
Web Form API
Chat Widget (Websocket)
Phone/IVR
API GATEWAY / LOAD BALANCER
CORE PROCESSING LAYER
MESSAGE QUEUE (e.g., Kafka)
TICKET ANALYSIS SERVICE
Classification (ML model)
Sentiment Analysis
Urgency Detection
AGENT MANAGEMENT SERVICE
Agent Availability Tracking
Skill Tagging (e.g., SQL, Spanish, Billing)
Performance Metrics
DYNAMIC ROUTING ENGINE
Routing Rules (Skill, Round Robin)
SLA Calculation
Load Balancing
(OPTIONAL) AI/ML MODEL TRAINING PLATFORM
PERSISTENCE LAYER
RELATIONAL DATABASE
(e.g., PestgreSQL for tickets)
NoSQL DB / REDIS
(for caching, real-time agent status)
ASSIGNMENT & WORKFORCE DELIVERY
ASSIGNED AGENTS
AGENT MOBILE APP / PUSH NOTIFICATIONS
MONITORING & REPORTING
GRAFANA DASHBOARD
ALERTS/LOGGING

Proposed Solution:

Project team shall fill the following information in the proposed solution.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Assigning support tickets manually is slow and can lead to mistakes and uneven workload. This causes delays and poor customer service. A simple and efficient system is needed to assign tickets quickly and fairly.
2.	Idea / Solution description	Create a system that automatically assigns tickets to available agents. It will distribute work evenly and help resolve issues faster.
3.	Novelty / Uniqueness	This project is unique because it uses smart rules to automatically assign tickets based on agent availability and workload. It ensures fair distribution of tasks and faster response, improving overall support efficiency compared to manual methods.
4.	Social Impact / Customer Satisfaction	The system improves customer satisfaction by providing faster responses and quicker issue resolution. It also reduces workload stress on support staff by distributing tasks fairly, leading to better service quality.

5.	Business Model (Revenue Model)	The system can be offered as a subscription-based service where companies pay a monthly or yearly fee to use the platform
6.	Scalability of the Solution	The system can easily handle more users and tickets as the business grows, without affecting performance.

Solution Architecture:

The solution architecture for the ticket assignment system is designed to provide an automated, scalable, and efficient approach to managing support operations. It integrates multiple components that work together to ensure smooth ticket flow from creation to resolution.

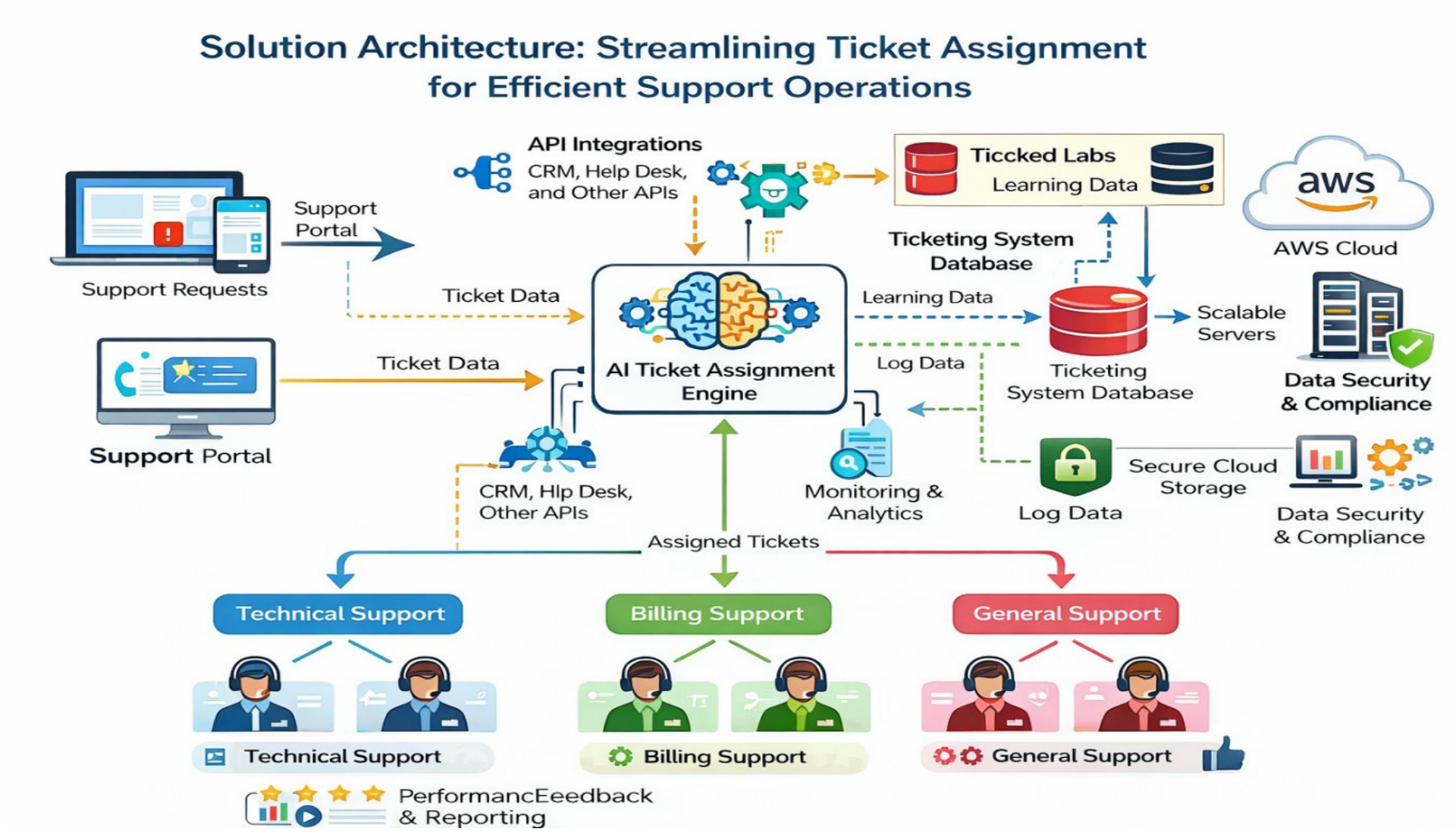
The system begins with the **User Interface Layer**, where customers or end users submit their support tickets through a web or application interface. This layer ensures easy interaction and captures all necessary ticket details such as issue type, description, and priority.

The submitted data is then processed in the **Application Layer**, which handles ticket validation, categorization, and prioritization. This layer ensures that the input data is accurate and structured properly before further processing.

Solution architecture bridges business challenges and technology solutions for **efficient ticket assignment**. Its goals are to:

- Implement an automated system to reduce manual effort and prioritize tickets.
- Define system structure, features, and behavior for stakeholders.
- Specify development phases, integration, and scalability requirements.

Example - Solution Architecture Diagram:



Problem – Solution Fit:

Organizations handling high volumes of tickets (IT support, customer service, ops teams) face:

1. Manual & Inefficient Assignment

- Tickets are assigned by team leads or first responders manually
- Leads to delays, especially during peak hours

2. Poor Skill Matching:

- Tickets often go to the wrong agent
- Causes reassignments → increased resolution time

3. Uneven Workload Distribution:

- Some agents are overloaded while others are underutilized
- Leads to burnout + inefficiency

4. Lack of Contextual Prioritization:

- Urgent tickets may not be recognized immediately
- SLA breaches and poor customer experience

5. Limited Visibility & Tracking:

- No real-time insights into who is working on what
- Hard to optimize or improve over time

Project Fit: Streamlining Ticket Assignment for Efficient Support Operations

CUSTOMER SEGMENT(S) — CS

- Businesses providing customer support
- Delays in ticket handling and resolution

CUSTOMER CONSTRAINTS — CC

- Limited staff for managing high ticket volume
- Time wasted in manual ticket assignment
- Difficulity in tracking ticket status

5. AVAILABLE SOLUTIONS — AS

- Manual assignment causing delays
- Basic ticket tracking systems not cefficient
- High-ticket volume overwhelms agents
- None are effective in automating and optmizing assignment

2. JOBS-TO-BE-DONE / PROBLEMS — JtBP

- Quicrfy assign tickets to available support agents
- Ensure tickets are handled efficiently
- Improve customer satisfaction

3. PROBLEM ROOT CAUSE — RC

- Manual ticket assignment, leads to slow response,
- workload imbalance, human errors
- Lack of smart classification delays assignments

7. BEHAVIOR — BH

- Currently, tickets are manuany saigned by managers
- High-ticket volume causes delays
- Managers find it hard to balance workload and track ticket satus

3. JOBS-TO-BE-DONE / PROBLEMS — JtBP

- Quickly assign tickets to available support agents
- Ensure tickets are handled efficiently
- Improve customer satisfaction

4. PROBLEM ROOT CAUSE — RC

- Currently, tickets are manually assigne
- High-ticket volume causes delays
- Managers find it hard to balance workloao

8. BEHAVIOR — BH

- Currently, tickets are manually assigned by managers
- High-ticket volume causes delays
- Managers find it hard to balance workload / and track ticket status

1. TRIGGERS — TR

- Frustration with slow responses
- Desire for quicker and efficient issue resolution
- Need to balance agent workload

9. YOUR SOLUTION — SL

- Automated ticket assignment based on agent availability and skills
- Categorizes and assigns tickets smartly and efficiently
- Real-time tracking for agents and customers
- Balanced workload distribution among agents

14. CHANNELS & BEHAVIOR — CH

- Centralized dashboard for agents and admin
- Mobile/ Web application for customers
- Email/ app notifications for instant alerts

2. EMOTIONS: BEFORE / AFTER — EM

- Before → Frustration from delayed reesolution vs. after?
- After → Relief and satisfaction with haaxv; efficient support

10. YOUR SOLUTION — SL

- Automated ticket assignment-based on agent availability and skills
- Categorizes and assigns tickets smartly and efficiently
- Balanced workload distribution among agents

6. MATIONS & BEHAVIOR — CH

- Centralized dashboard for agents and admin
- Mobile/ Web application for customers
- Email/ app notifications for instant alerts

3. EMOTIONS: BEFORE / AFTER — EM

- Before → Frustration from delayed responses
- After → Relief and satisfaction with haaxv; efficient support

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

Model Performance Testing: Ticket Assignment Optimization

1. Metrics Overview Table

Category	Metric	Definition	Goal
🧠 **Model Accuracy**	Classification Accuracy	% of correctly classified tickets (category/priority)	Higher is better
🧠 **Model Accuracy**	Precision	Correct positive predictions / total predicted positives	Reduce false alarms
🧠 **Model Accuracy**	Recall	Correct positives / actual positives	Catch all critical tickets
🧠 **Model Accuracy**	F1 Score	Balance between precision and recall	Overall model quality
⚙ **Assignment Quality**	First-Time Right Assignment (FTRA)	% tickets assigned correctly without reassignment	Maximize
⚙ **Assignment Quality**	Reassignment Rate	% tickets reassigned after initial routing	Minimize
⚙ **Assignment Quality**	Skill Match Score	Degree of match between ticket type and agent skill	Maximize
⏱ **Efficiency**	Avg Assignment Time	Time taken to assign ticket to agent	Minimize
⏱ **Efficiency**	Time to First Response (TTFR)	Time until agent responds	Minimize
📈 **Operations**	SLA Compliance Rate	% tickets resolved within SLA	Maximize
⚖ **Load Balance**	Agent Utilization Rate	Work distribution across agents	Balanced distribution
⚖ **Load Balance**	Workload Variance	Difference in tickets per agent	Minimize

Model performance testing in this project focuses on evaluating the efficiency and accuracy of the ticket assignment system.

The main objective is to ensure that tickets are correctly categorized and assigned to the appropriate support agents with minimal delay.

The system is tested based on key performance metrics such as assignment accuracy, response time, and error rate.

Assignment accuracy measures how correctly the system assigns tickets to the right agents based on issue type, priority, and availability. A high accuracy indicates that the system is functioning effectively.

The testing process also includes validating the assignment logic and ensuring that all predefined rules are applied correctly. Screenshots and results are recorded to verify system performance.

Solution Architecture:

⍰ High-Level Architecture Overview

Layer	Component	Responsibility
□ Input Layer	Ticket Intake System	Captures tickets from email, web, chat, API
□ Processing Layer	Preprocessing & NLP Engine	Cleans, extracts features, classifies tickets
□ Intelligence Layer	ML Model + Rules Engine	Predicts category, priority, and assigns agent
□ Routing Layer	Assignment Engine	Applies skill matching + load balancing
□ Execution Layer	Agent Dashboard	Displays assigned tickets to agents
📊 Monitoring Layer	Analytics & Logging System	Tracks performance, SLA, and system health

5. PROJECT PLANNING & SCHEDULING:

1. Key Definitions:

🎯 Objective

To define a structured plan and timeline for developing, testing, and deploying an intelligent ticket assignment system that improves routing efficiency, reduces manual intervention, and enhances SLA compliance.

🗓 1. Project Planning Overview:

Phase	Description	Key Output
Phase 1	Requirement Gathering & Analysis	Problem statement, scope, SRS
Phase 2	System Design	Architecture, data flow, module design
Phase 3	Data Collection & Preparation	Cleaned and labeled ticket dataset
Phase 4	Model Development	ML model for classification & prioritization
Phase 5	Rules Engine Development	SLA, VIP, and routing rules
Phase 6	System Integration	ML model + assignment engine integration
Phase 7	Testing & Validation	Performance test reports
Phase 8	Deployment	Production-ready system
Phase 9	Monitoring & Optimization	Dashboards and continuous improvement

Sprint 1 — Data Handling Improve how support tickets are **collected, processed, and assigned**, so they reach the right team faster with fewer manual steps and errors.

2.Sprint 1Focus Areas:

1. Data Standardization:

Ensure incoming ticket data is clean and structured:

- Normalize fields (category, priority, customer type)
- Enforce required fields at intake
- Remove duplicate or inconsistent formats

---

2. Ticket Intake Pipeline Improvements:

- Improve ingestion from email/forms/API/chat
- Validate and sanitize incoming ticket data
- Add basic enrichment (e.g., customer ID mapping, product tagging)

---

3. Rule-Based Assignment Engine (MVP):

Implement initial routing logic:

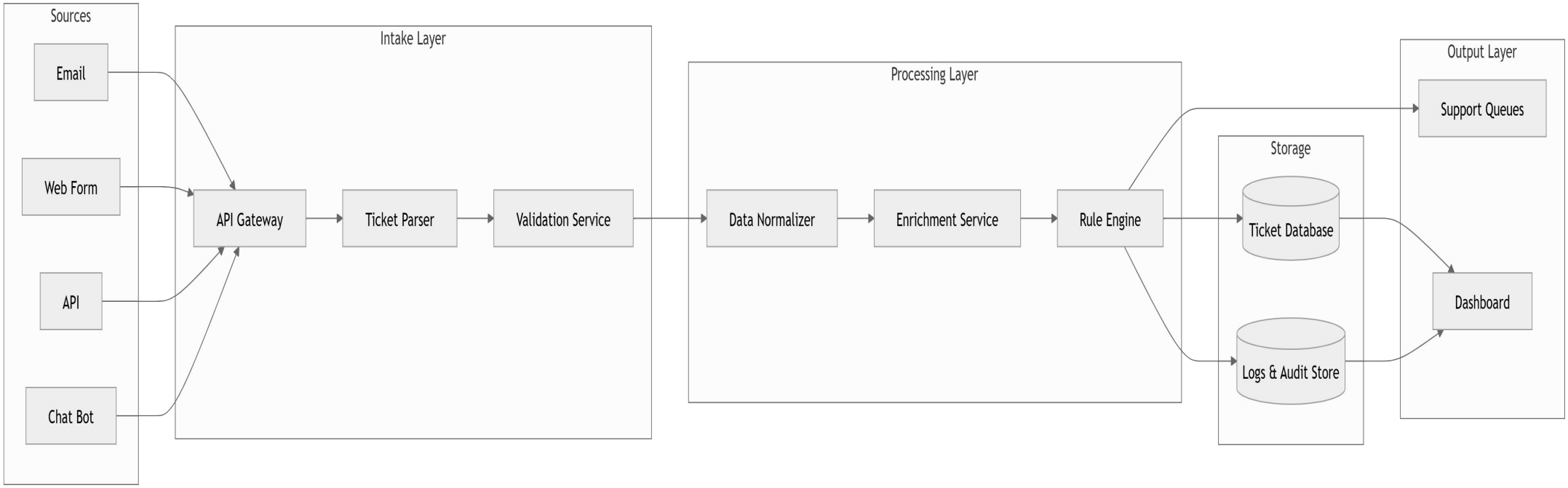
- If `category = billing` → Billing team
- If `priority = high` → Escalation queue
- If `product = X` → Product X support group

4. Data Logging & Traceability:

- Log assignment decisions
- Track changes in ticket status
- Store assignment history for audits/debugging

5. Basic Dashboard/Monitoring (Optional MVP):

- Number of tickets by category
- Assignment time average
- Misrouted tickets count



3. Sprint 2 — Visualization & Reporting

Sprint 2 focuses on visualizing the processed employee data and building dashboards and story reports for stakeholders.`Enable` stakeholders (HR, managers, leadership) to **visually explore employee data** to gain insights into workforce composition, performance, and trends for better decision-making.

🎯 Key Objectives:

1. Unified Employee Data Model:

- Consolidate data from HR systems, payroll, attendance, and performance tools
- Standardize fields like:
  - Department
  - Role
  - Tenure
  - Performance rating
  - Location

---

2. Data Processing Layer:

- Clean and normalize employee datasets
- Handle missing or inconsistent values
- Aggregate data for analytics (monthly, quarterly views)

---

3. Visualization Dashboard (Core Feature):

Create interactive dashboards with:

- Employee distribution by department
- Headcount trends over time

- Attrition rate analysis
- Performance rating distribution
- Location-wise workforce mapping

---

4. Filters & Drilldowns:

- Filter by department, role, location, tenure
- Drill down from company → department → team → individual
- Time-based filtering (monthly/quarterly/yearly)

---

5. Role-Based Access Control:

- HR: full access
- Managers: team-level data only
- Leadership: aggregated insights

📦 Deliverables:

- Clean employee analytics dataset
- Visualization dashboard (MVP)
- Filter + drilldown functionality
- Role-based access control setup

📊 Success Metrics:

- ⏱️ Reduced time to generate workforce reports

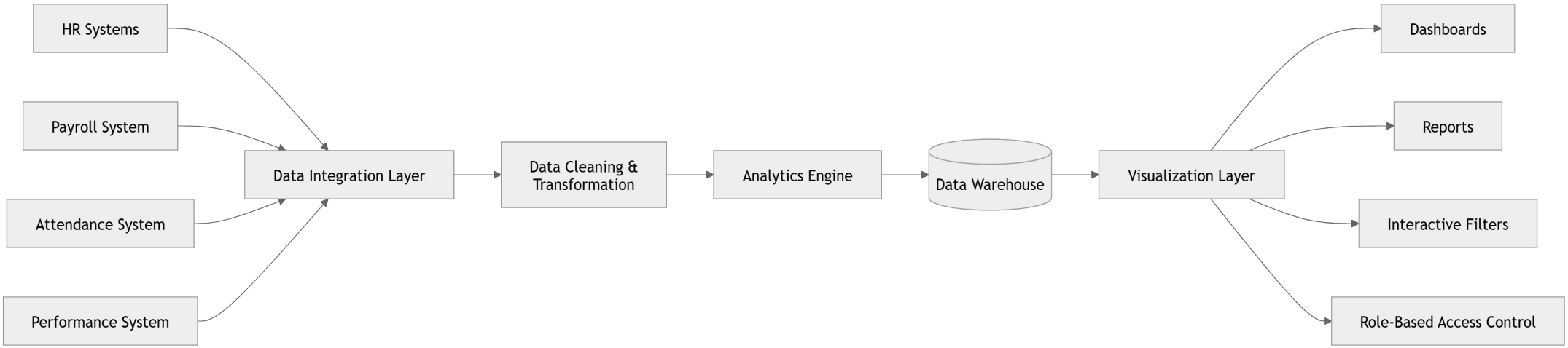
- 📈 Increased dashboard usage by stakeholders
- 🧠 Improved decision-making speed
- 📉 Reduced dependency on manual Excel reports

⚠ Risks / Challenges:

- Inconsistent HR data across systems
- Data privacy and access control complexity
- Performance issues with large datasets
- Misinterpretation of visual insights without context

🔜 Future Enhancements (Next Epics):

- Predictive analytics (attrition risk, hiring needs)
- AI-driven insights ("why attrition increased?")
- Real-time dashboard updates
- Benchmarking against industry data



Epic Goal:

Build a **centralized, interactive dashboard** that provides real-time visibility into key business metrics, enabling faster decision-making and improved operational control.

Key Objectives:

1. Unified KPI Layer:

Define and standardize key metrics:

- Operational KPIs (tickets, resolution time, SLA)
- Employee KPIs (headcount, attrition, performance)
- Business KPIs (revenue, growth, utilization)

---

2. Data Aggregation Layer:

- Collect data from multiple systems
- Build a centralized analytics-ready dataset
- Ensure data freshness (batch or near real-time)

---

3. Dashboard UI Development:

Create an interactive dashboard with:

- KPI summary cards (top-level metrics)
- Trend charts (daily/weekly/monthly)
- Breakdown views (department, team, region)
- Drill-down capability

---

4. Filters & Interactivity:

- Time range selector
- Department/team filters
- Role-based filtering
- Dynamic chart updates

---

5. Role-Based Views:

- Executives → high-level KPIs only
- Managers → team-level insights
- Analysts → detailed raw + aggregated data

Velocity Calculation (Project-Based):

Velocity in Agile = **amount of work completed in a sprint**, usually measured in **Story Points**.For your project (Epics 3 & 4: *Employee Data Visualization + Dashboard*), here's how velocity would be calculated in a realistic way.

Step 1: Break Work into Stories (Example)

📊 Epic 3: Employee Data Visualization:

Story	Description	Points
US1	Build unified employee data model	8

Story	Description	Points
US2	Data cleaning & normalization pipeline	5
US3	Department-wise visualization charts	5
US4	Attrition trend analysis chart	3
US5	Role-based access control	5

✔ Epic 3 Total = 26 Story Points

📊 Epic 4: Dashboard:

Story	Description	Points
US6	KPI definition layer setup	5
US7	Data aggregation pipeline	8
US8	Dashboard UI (cards + charts)	8
US9	Filters & drilldowns	5
US10	Role-based dashboard views	5

✔ Epic 4 Total = 31 Story Points

📦 Total Project Backlog:

👉 26 + 31 = **57 Story Points**

🚀 Step 2: Sprint Velocity Example:

Assume your team completes work like this:

Sprint 1

- Completed: US1, US2, US6
  👉 8 + 5 + 5 = **18 points**

Sprint 2

- Completed: US3, US4, US7
  👉 5 + 3 + 8 = **16 points**

Sprint 3

- Completed: US5, US8
  👉 5 + 8 = **13 points**

Sprint 4

- Completed: US9, US10
  👉 5 + 5 = **10 points**

📊 Step 3: Velocity Calculation:

Sprint Velocities:

- Sprint 1 → 18
- Sprint 2 → 16

- Sprint 3 → 13
- Sprint 4 → 10

📌 Average Velocity:

(18+16+13+10)/4=57/4=14.25(18 + 16 + 13 + 10) / 4 = 57 / 4 = 14.25(18+16+13+10)/4=57/4=14.25

🎯 **Final Velocity ≈ 14 Story Points per Sprint**

📉 Interpretation:

- Team capacity ≈ **14 points per sprint**
- Total work = **57 points**
  -

  Estimated completion time:

  - 57/14≈4.1 sprints57 / 14 ≈ 4.1 \text{ sprints}57/14≈4.1 sprints

👉 So project finishes in ~**4 to 5 sprints**

📊 Velocity Chart (Simple View):

```
Sprint 1 | ■■■■■■■■■■■■■■■■■■■■■■ 18
Sprint 2 | ■■■■■■■■■■■■■■■■■■■■■■    16
Sprint 3 | ■■■■■■■■■■■■■■■■■■■
```

Below is the **full story point breakdown** for your project covering:

- Epic 3: Employee Data Visualization
- Epic 4: Dashboard

Story ID	Feature	Story Points
US1	Unified employee data model	8
US2	Data cleaning & normalization pipeline	5
US3	Department-wise visualization	5
US4	Attrition trend analysis	3
US5	Role-based access control	5

servicenow All Favorites History Table - operations related Search

Access Control
u_operations_related

Type	record	Application	Global
Operation	read	Active	✓
Decision Type	Allow If	Advanced	
Admin overrides	✓		
Protection policy	-- None --		
Name	u_operations_related		
Description	Default access control on u_operations_related		
Applies To	(empty)		

Conditions

Requires role

Role

Gowthami kota

Streamlining Ticket Assignment in ServiceNow | Step-by-Step implementation

servicenow All Favorites History Admin User - New Record Search

User
New record

Submit

To set up the User's password, save the record and then click Set Password.

User ID	katherine pierce	Email	kather
First name	katherine	Language	-- None --
Last name	pierce	Calendar integration	Outlook
Title		Time zone	System (America/Los_Angeles)
Department		Date format	System (yyyy-MM-dd)
Password needs reset	☐	Business phone	
Locked out	☐	Mobile phone	
Active	☑	Photo	Click to add...
Web service access only	☐		
Internal Integration User	☐		

Submit

Related Links

Gowthami kota

1:31 / 26:51

servicenow All Favorites History Access Control - New Record Search

Access Control
New record

Submit

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type	record	Application	Global
* Operation	write	Active	☑
Decision Type	Allow If	Advanced	☐
Admin overrides	☑		
Protection policy	-- None --		
* Name	operations related [u_operations_related]		priority
Description			

Applies To No. of records matching the condition: 0

Add Filter Condition | Add OR Clause

-- choose field -- | -- oper -- | -- value --

Conditions

Gowthami kota

Streamlining Ticket Assignment in ServiceNow | Step-by-Step implementation

Workflow Studio | regarding certificate Flow • Global | regarding platform Flow • Global

regarding platform Inactive

View: Test Debug Activate Save

TRIGGER

operations related Created or Updated where (issue is unable to login to platform)

Trigger Created or Updated

Table

* Table operations related [u_operations...

Condition All of these conditions must be met

issue | is | unable to login | OR | AND

or

New Criteria

Run Trigger Once

Advanced Options

Delete | Cancel | Done

Data Collapse All

Flow Variables

Trigger - Record Created or Updated

operations related Record	Record
Changed Fields	Array.Object
operations related Table	Table
Run Start Time UTC	Date/Time
Run Start Date/Time	Date/Time

Share

Gowthami kota

javascript:void(0) | Application: Global

✅ Advantages:

1. ⚡ Faster Ticket Resolution:

- Automated or rule-based assignment routes tickets quickly to the right team
- Reduces waiting time in queues

---

2. 🎯 Improved Accuracy in Routing:

- Tickets are assigned based on **rules, categories, or metadata**
- Minimizes human errors and misclassification

---

3. 🔄 Reduced Manual Effort:

- Eliminates need for support agents to manually triage tickets
- Frees up agents for higher-value tasks (resolution instead of sorting)

---

4. 📈 Better SLA Compliance:

- Faster assignment improves response time and resolution time
- Helps meet SLA targets consistently

---

5. 📊 Increased Operational Visibility:

- Logs and tracking provide insights into:
  - Ticket flow
  - Bottlenecks
  - Team workload distribution

---

6. ⚖️ Balanced Workload Distribution:

- Tickets can be routed evenly across teams or based on capacity
- Prevents overload on specific support groups

---

7. 🧠 Foundation for Automation & AI:

- Enables future upgrades like:
    - ML-based classification
    - Predictive routing
    - Smart prioritization

✕ Disadvantages:

1. ⚠ Dependency on Data Quality:

- Incorrect or incomplete ticket data leads to wrong assignment
- Bad input = bad routing decisions

---

2. ❓ Complex Rule Management:

- Rule-based systems become difficult to maintain as conditions grow
- Conflicts between rules can cause misrouting

---

3. 📉 Limited Flexibility (MVP Systems):

- Static rules may not handle edge cases well
- New scenarios require frequent updates

---

4. ⏳ Initial Setup Effort:

- Requires effort to design workflows, rules, and data pipelines
- Needs coordination between support + engineering teams

---

5. 🔐 Risk of Incorrect Assignment:

- Misrouted tickets can delay resolution
- May require re-routing and create overhead

---

6. 📊 Scalability Challenges:

- High ticket volume may slow down processing if system is not optimized
- Needs scalable architecture for large organizations

---

7. 🧠 Lack of Context Awareness (Rule-Based Systems):

- Simple rules cannot understand complex ticket context
- Advanced AI/ML needed for smarter routing

CONCLUSION:

In conclusion, the **Streamlining Ticket Assignment for Efficient Support Operations** project significantly improves the efficiency, accuracy, and speed of handling customer support requests within an Agile engineering environment.By introducing a structured and automated approach to ticket ingestion, validation, enrichment, and rule-based assignment, the system reduces manual intervention and ensures that each ticket is routed to the appropriate support team in a timely manner. This leads to faster response times, improved SLA compliance, and better customer satisfaction.

The implementation of standardized data handling and tracking mechanisms also enhances transparency and traceability across the ticket lifecycle. Additionally, the solution provides a strong foundation for future enhancements such as machine learning-based classification, intelligent routing, and predictive support operations.Although challenges such as data quality dependency, rule maintenance complexity, and scalability considerations exist, these can be addressed through continuous iteration in Agile sprints and system optimization.

Overall, this project demonstrates how Agile principles combined with data-driven engineering solutions can transform traditional support workflows into a more efficient, scalable, and intelligent ticket management system.

Future Scope:

The future scope of the **ticket assignment system** focuses on making the support process more **intelligent, automated, and scalable** beyond basic rule-based routing.

---

1. 🤖 AI-Based Ticket Classification:

In the future, Machine Learning models can be used to automatically analyze incoming tickets and classify them based on category, priority, and urgency. This reduces manual tagging and improves accuracy by learning from historical ticket data.

---

2. 🧠 NLP-Based Understanding:

Natural Language Processing (NLP) can help the system understand the actual meaning of customer messages. It can detect sentiment, urgency, and intent, allowing smarter and more context-aware ticket routing.

---

3. ⚡ Real-Time Processing:

Instead of batch processing or delayed assignment, future systems can use real-time event processing so that tickets are assigned instantly as soon as they are created. This improves response time and reduces waiting queues.

---

4. 📊 Advanced Analytics Dashboard:

A more powerful dashboard can provide insights like SLA performance, agent efficiency, ticket resolution trends, and workload distribution. It can also support predictive analytics to forecast future ticket volume.

---

5. 🔄 Smart Workload Balancing:

Future systems can automatically distribute tickets based on agent availability, skills, and workload. This ensures fair distribution and prevents overloading specific support teams.

---

6. 🔐 Enhanced Security:

The system can be upgraded with stronger security mechanisms like role-based and attribute-based access control. This ensures that sensitive customer data is protected and only accessible to authorized users.

---

7. 🌐 Integration with Other Systems:

The ticketing system can be integrated with CRM tools, chatbots, and communication platforms. This creates a unified support ecosystem where all customer interactions are managed in one place.

---

8. 📱 Omnichannel Support:

Future improvements can allow tickets to be created from multiple channels like email, chat, mobile apps, and social media while maintaining a single unified system.

THANK YOU…